

Step 4 Validation Protocol: Independent Temperature and Field Trends

1 Purpose of the test

Step 3 showed that a dynamic dimerisation/phonon coordinate can produce a nonlinear response that is distinct from a purely scalar bright-dark mixing. Step 4 tests the trend-level interpretation needed before assigning a measured feature to spin-Peierls physics. The control is:

$$\delta(T, B) \rightarrow 0 \quad \text{above the spin-Peierls transition,} \quad (1)$$

while the local spin correlation proxy remains finite,

$$C_1(T, B) \neq 0 \quad \text{in both uniform and dimerised regimes.} \quad (2)$$

The second condition is essential: if C_1 were forced to zero above the transition, the test would no longer distinguish a lattice order parameter from a uniform spin-correlation channel.

2 Physical model

The Hamiltonian is the Step 3 model,

$$H_{\text{mix}} = V_{BD}^{\text{static}} L_{BD} + g_Q^{\text{eff}} Q L_{BD}, \quad (3)$$

with

$$V_{BD}^{\text{static}} = V_0 + \lambda_\delta \delta(T, B) + \lambda_C C_1(T, B). \quad (4)$$

The dynamic lattice coupling is scaled by the spin-Peierls order parameter,

$$g_Q^{\text{eff}} = g_{Q0} \frac{\delta(T, B)}{\delta(T_{\text{ref}}, B_{\text{ref}})}. \quad (5)$$

Thus the dynamic lattice channel turns off when the dimerisation turns off.

3 Trend definitions

The spin-Peierls transition temperature is represented phenomenologically by

$$T_{\text{SP}}(B) = \max [0, T_{\text{SP},0}(1 - \alpha_B B^2)]. \quad (6)$$

The dimerisation is

$$\delta(T, B) = \begin{cases} \delta_0 (1 - T/T_{\text{SP}}(B))^\beta, & T < T_{\text{SP}}(B), \\ 0, & T \geq T_{\text{SP}}(B). \end{cases} \quad (7)$$

The spin-correlation proxy is not an order parameter. It is kept finite above the transition and varies smoothly with temperature and field.

4 Scenarios

Scenario	Regime	Lattice channel	Purpose
<code>lowT_lattice_dynamic</code>	low T , low B	on	spin-Peierls reference
<code>highT_lattice_off_C1_finite</code>	high T	off	uniform chain with finite C_1
<code>highB_lattice_suppressed</code>	low T , high B	off	field-suppressed order
<code>highT_C1_static</code>	high T	off	persistent scalar C_1 signal

5 Acceptance criteria

A lattice-controlled feature should be large in the low-temperature spin-Peierls case and suppressed in both the high-temperature and field-suppressed controls. A feature that remains large in the `highT_C1_static` case is compatible with a spin-correlation channel, but it should not be interpreted uniquely as evidence for the dimerised phase.

6 Requested outputs

For each scenario, save:

- the rephasing real spectrum PDF;
- the pathway-resolved complex response arrays;
- the rephasing, unrephasing, and 1Q total components;
- text and JSON summaries containing T , B , T_{SP} , δ , C_1 , g_Q^{eff} , and V_{BD}^{static} .

The analysis should compare finite-window integrals, full-spectrum differences, and trend metrics across the four scenarios.

7 Numerical results

The four Step 4 scenarios were generated at $\eta = 0.001$ with a 120×120 frequency grid, $n_{\text{bosons}} = 3$, and $\omega_Q = 0.035$ eV. The finite-window half-width used for the aggregate metrics is 0.025 eV.

Scenario	δ	C_1	g_Q^{eff}	V_{BD}^{static}	A_{cross}
Low- T lattice dynamic	0.007071	-0.378667	0.0100	0.0200	12.554241
High- T lattice off	0	-0.329471	0	0.0100	10.919127
High- B lattice suppressed	0	-0.346199	0	0.0100	10.919127
High- T C_1 static	0	-0.329471	0	0.0200	16.266623

The low-temperature lattice reference has finite δ and finite g_Q^{eff} . The high-temperature and high-field lattice-off controls both have $\delta = 0$ and $g_Q^{\text{eff}} = 0$, while C_1 remains finite. Because these two controls also have the same $V_{BD}^{\text{static}} = 0.01$, their spectra are identical in this minimal model. The full-spectrum relative maximum difference between them is exactly zero at numerical precision.

The low-temperature lattice spectrum differs strongly from the high-temperature lattice-off control. The full-spectrum relative maximum difference is 6.76×10^{-1} , and the main cross-window

area changes from 12.554241 to 10.919127. This confirms that the implemented lattice channel follows the spin-Peierls trend.

The high-temperature C_1 static control is the key interpretive warning. It keeps $\delta = 0$ and $g_Q^{\text{eff}} = 0$, but uses finite C_1 to restore $V_{BD}^{\text{static}} = 0.02$. Its main cross-window area is 16.266623, larger than both lattice-off controls. The full-spectrum relative maximum difference between the high-temperature lattice-off control and this C_1 static control is 5.98×10^{-1} .

8 Validation outcome

The trend-control validation passes. The model now contains two independent trend channels: a dimerisation/lattice channel that vanishes above the spin-Peierls transition or under field suppression, and a spin-correlation proxy that remains finite in the uniform regime.

The conclusion is therefore sharper than in Step 3. A feature that disappears when $\delta(T, B)$ and g_Q^{eff} vanish is compatible with a spin-Peierls/lattice-controlled mechanism. A feature that persists in the high-temperature C_1 static control, where $\delta = 0$, cannot be assigned uniquely to the dimerised phase. It may instead reflect a scalar local spin-correlation contribution to bright-dark mixing.

9 Conclusions and next step

Step 4 provides the minimal trend-level filter needed before connecting the model to CuGeO3 data. The lattice channel should be judged by whether it tracks $\delta(T, B)$, not merely by whether the signal is present at low temperature. The finite- C_1 control shows that a nonlinear feature can remain strong even when the spin-Peierls order parameter is zero.

The next useful calculation is to turn this discrete control into a scan:

- sweep T across T_{SP} at fixed low field and track selected window integrals;
- sweep B at fixed low temperature to check field suppression of $\delta(T, B)$;
- compare lattice-dynamic and C_1 -static channels using the same plotted observables rather than only full spectra;
- only after that, add a momentum-sector or spinon-inspired treatment if the trend-level signatures remain distinct.